MicroSense AI-Cam Sample Analysis Report

Report Generated:
Jun 08, 2026 02:26 PM

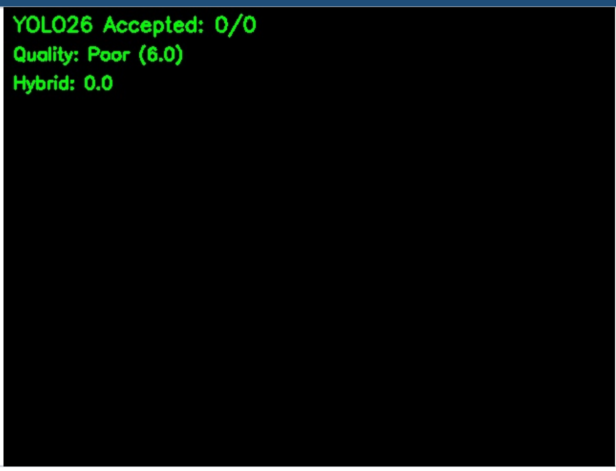
1. SAMPLE INFORMATION

Sample ID	MSC-023
Sample Source	Hardware USB Microscope
Chamber Volume	50 mL
Analysis Date	Jun 08, 2026 08:55 AM
Detection Model	YOLO26n + Hybrid Filter
File Type	image
Sample Notes	Captured using ESP32-triggered ha...

2. RESULTS SUMMARY

Detected Candidates	0 particles
Estimated Concentration	0 particles/L
MSMI Score	0.00 / 1.00
Confidence Score	0.00 / 1.00
Hybrid Validation	0.00 / 1.00
Image Quality	Poor
Quality Warning	Image may be blurry. Image is too...
Monitoring Risk	Low
Recommendation	Low monitoring concern. Continue...

3. PROCESSED IMAGE EVIDENCE



Detected Candidate Particles: 0

- Accepted candidates (0)
- Rejected candidates (0)

AI Screening Method

- YOLO26n detection identifies visible candidate particles from the optical image.
- Hybrid validation checks shape, contrast, edge clarity, brightness, and particle size consistency.
- MSMI and risk level provide a preliminary monitoring interpretation for the sample.

4. INTERPRETATION AND LIMITATION

The uploaded water sample contains visually detected microplastic-like particle candidates. The result is calculated from optical image evidence, accepted detections, estimated concentration, image quality, and hybrid validation. This report can support rapid preliminary screening, sample comparison, and routine monitoring decisions.

Disclaimer: MicroSense AI-Cam does not perform chemical polymer identification and does not replace FTIR spectroscopy, Raman spectroscopy, or certified laboratory analysis. Confirmatory testing by an accredited laboratory is required for material identification, official reporting, or regulatory compliance.

5. MONITORING NOTES

Recommended use: compare repeated samples from the same source, track changes in accepted candidate counts, review MSMI trends, and preserve the generated PDF as supporting screening documentation. For high-risk or unusual readings, repeat imaging under consistent lighting and proceed with confirmatory analysis.

6. OUTPUT CHECKLIST

Screening Completed

AI-assisted optical analysis completed for the uploaded sample.

Report Archived

PDF can be stored with sample history for later review.

Further Action

Repeat imaging or confirm through laboratory analysis when required.